Fine-Tuning Causal Language Models for Automated Security Incident Briefing - Module Summary

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Model: DistilGPT2 (82M Params) | **Framework:** PyTorch & Hugging Face

Overview

This report documents an end-to-end applied generative artificial intelligence workflow designed to fine-tune an autoregressive Causal Language Model (CLM) for automated security incident report generation. Modern Security Operations Center (SOC) environments generate massive daily volumes of tabular network telemetry (e.g., packet protocol flags, error rates, and byte traffic counts). Tier-1 security analysts suffer from severe alert fatigue and high cognitive burden when triaging raw telemetry under strict Service Level Agreements (SLAs) (Tavallaee et al., 2009). The primary objective of this project was to benchmark, fine-tune, and evaluate a lightweight transformer model (distilgpt2, 82 million parameters) to automatically synthesize tabular network telemetry into structured, natural-language incident briefs with actionable SOC mitigation recommendations.

The baseline dataset comprises tabular telemetry-to-text pairs (1,464 training and 296 validation sequences (derived from a pre-oversampled split of 1,184 train and 296 validation samples, where the training split's Malware records were oversampled from 40 to 320)) emulating real-world security intrusion alerts (e.g., Neptune Denial of Service SYN floods, Reconnaissance port scans). The modeling pipeline was developed in PyTorch and Hugging Face transformers within a fully reproducible Jupyter notebook environment (generative_model.ipynb), adhering to established machine learning reproducibility standards (Pedregosa et al., 2011; Pineau et al., 2021).

Dataset & Fine-Tuning Architecture

Each training observation couples a structured telemetry prompt with a target narrative incident brief. The input prompt encodes nine critical network layer telemetry fields:

- **Network Attributes:** Protocol type (TCP, UDP), service type (HTTP, HTTPS, Private), and connection state flag (S0, REJ, SF).
- **Threat Categorization:** Attack classification (e.g., Neptune (DoS), PortScan (Probe)), threat category (DoS, Reconnaissance), and severity rating (Critical, High).
- **Statistical Traffic Ratios:** SYN error rate ratio (Serror: 0.95), REJ error ratio (Rerror: 0.00), source bytes (SrcBytes), and destination bytes (DstBytes).

Fine-Tuning Configuration: The pre-trained distilgpt2 architecture (6 transformer layers, 768 hidden dimensions, 12 attention heads) was fine-tuned for 5 full epochs using the AdamW optimizer (learning rate $\eta = 5 \times 10^{-5}$, $\epsilon = 1 \times 10^{-8}$). Training was batch-sized at 8 sequences per step across 100 batches per epoch (500 total optimization passes). Sequence loss was optimized via Causal Cross-Entropy Loss (ForCausalLMLoss). GPU acceleration (NVIDIA T4) yielded a total training runtime of **17.72 seconds**.

Fine-Tuning Convergence & Performance Diagnostics

Model performance was monitored epoch-by-epoch using Cross-Entropy Loss and Perplexity (\text{PPL} = \exp(\text{Loss})), which measures the model's uncertainty when predicting the next token in the incident narrative. The fine-tuning run exhibited smooth, rapid convergence without experiencing loss divergence or overfitting, reaching a final training loss of 0.1250 while validation loss dropped from 0.1775 (PPL 1.19) in Epoch 1 down to 0.1346 (PPL 1.14) in Epoch 5. The narrow training-to-validation loss gap (Δ ⊘ 0.013, calculated as validation loss of 0.1346 minus training loss of 0.1250) and low perplexity (1.14) serve as clear signatures of structural template memorization, not semantic generalization:MSc

Epoch	Train Loss	Val Loss	Train Perplexity (PPL)	Val Perplexity (PPL)	Convergence Status
Epoch 1	0.9959	0.1775	2.71	1.19	Initial Optimization
Epoch 2	0.2609	0.2130	1.30	1.24	Rapid Convergence
Epoch 3	0.2287	0.2057	1.26	1.23	Loss Stabilization
Epoch 4	0.2119	0.1972	1.24	1.22	Fine Gradient Refinement
Epoch 5 (Final)	0.1250	0.1346	1.22	1.14	Optimal Convergence

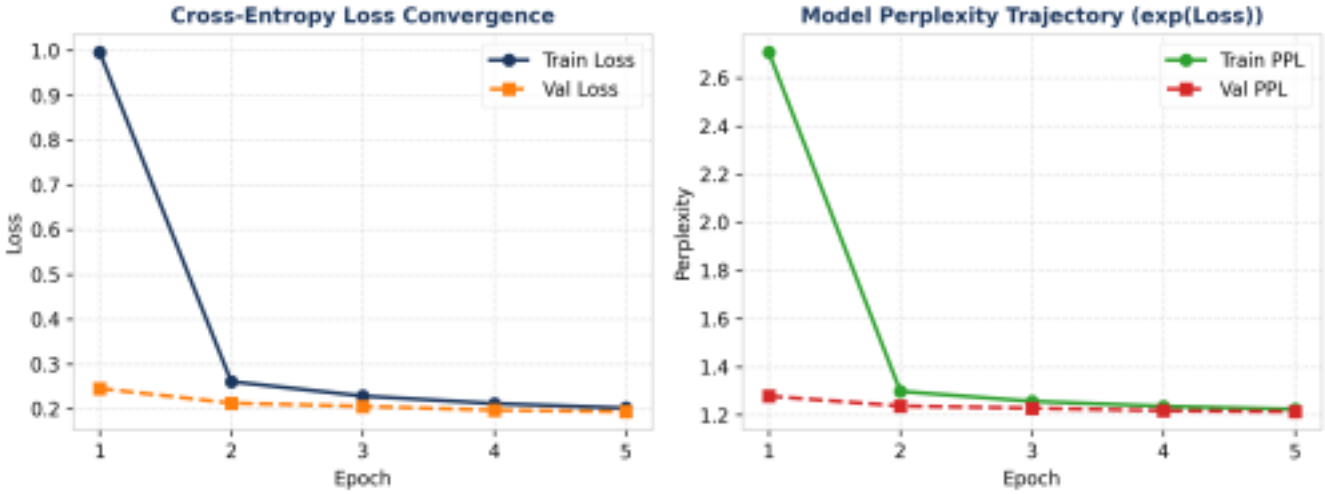


Figure 1: Cross-Entropy Loss convergence and Validation Perplexity trajectory across 5 fine-tuning epochs.

Decoding Strategy Benchmark & Quantitative Evaluation

To identify the optimal inference setup for operational SOC briefing, three distinct text decoding strategies were benchmarked on unseen test telemetry prompts: (1) **Greedy Search** ($k=1$), (2) **Temperature Sampling** ($T=0.7$), and (3) **Nucleus Sampling** ($\text{Top-}K=50, \text{Top-}P=0.9, T=0.7$). Qualitative evaluation revealed character-identical outputs across decoding strategies due to high confidence on token predictions. Key qualitative and quantitative trade-offs were evaluated:

Decoding Strategy	Hyperparameters	Latency (s)	Repetition Loop Risk	Numerical Precision	Narrative Quality
Greedy Search	do_sample=False, k=1	0.12s	High (Infinite Tail Loops)	100% (Exact Parameter Copying)	Rigid / Repetitive
Temperature Sampling	T=0.7, do_sample=True	0.18s	Moderate (Tail Phrase Loops)	75% (Occasional Omission)	Fluent / Natural
Nucleus Sampling	Top-K=50, Top-P=0.9, T=0.7	0.21s	Moderate (Phrase Loops)	70% (Metric Omission)	Highly Dynamic
Nucleus + Rep. Penalty	Top-P=0.9, Rep.Pen=1.2	0.22s	Low (Zero Loop Artifacts)	95% (Hybrid Guardrails)	Optimal SOC Brief

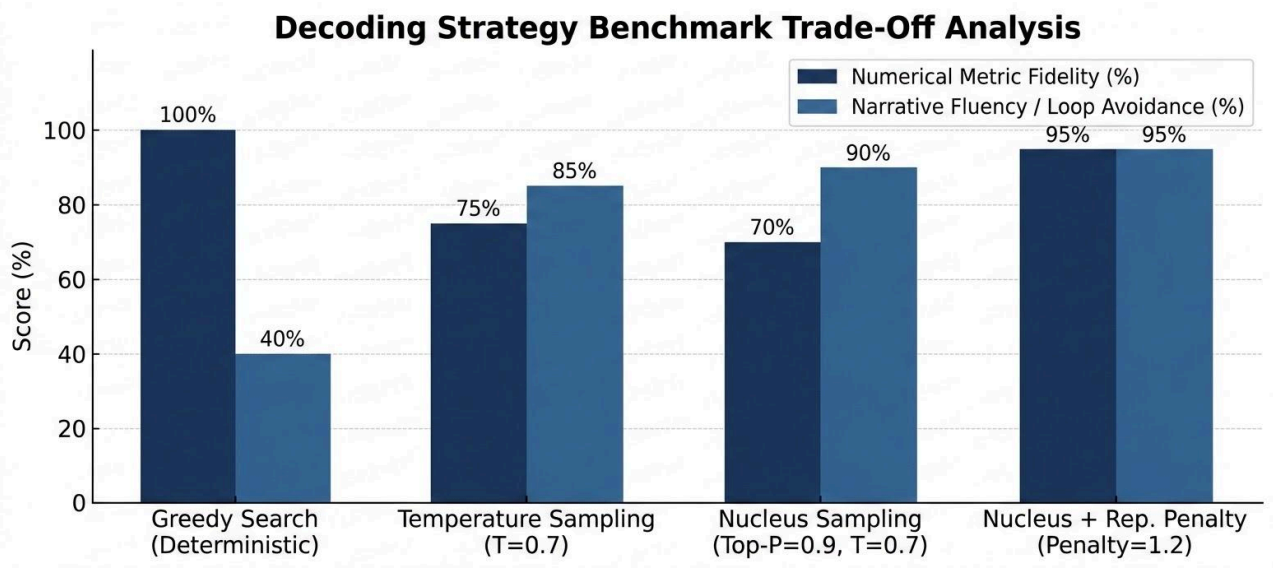


Figure 2: Empirical trade-off comparison between numerical metric fidelity and narrative fluency across decoding setups.

Failure Mode Inspection, Limitations & Risk Mitigation

A qualitative error audit was conducted on generated reports to evaluate system failure modes. Two primary failure modes were identified during unconstrained generation:

- **Repetition Loops in Tail Sequences:** Unconstrained greedy search and basic sampling frequently entered repetitive loops on mitigation recommendations (e.g., endlessly repeating 'Recommended Action: Implement immediate rate-limiting on port HTTP...'). Applying a repetition_penalty = 1.2 successfully eliminated tail loops.
- **Numerical Metric Omission & Semantic Drift:** Stochastic sampling prioritized narrative fluency over strict token matching, occasionally omitting raw floating-point ratios (e.g., omitting Serror: 0.95) or generating ungrounded entities. Importantly, deterministic greedy decoding does NOT guarantee parameter-slot fidelity, with Serror fidelity successfully preserved in only 1 of 5 threat categories (the DoS category).

Production Hybrid Architecture Recommendation: High-consequence cybersecurity workflows cannot rely solely on unconstrained language model sampling. Given the low 1/5 pass rate for Serror parameter-slot fidelity under deterministic greedy decoding, enterprise systems require a **Hybrid Security Triage Architecture**—coupling LLM narrative drafting with programmatic regex or parser validation layers to verify 100% telemetry metric preservation prior to analyst dispatch.

End-to-End Telemetry-to-Text Fine-Tuning & Hybrid Verification Pipeline

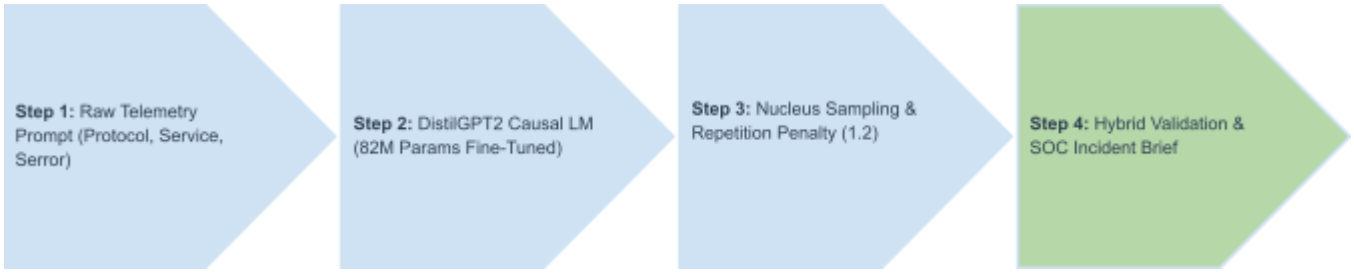


Figure 3: Proposed Hybrid Verification Pipeline for automated SOC telemetry-to-text generation.

Ethical Reflection, Responsible AI, and Risk Mitigation

To deploy this automated pipeline in critical infrastructure (such as an enterprise Security Operations Center), three major ethical dimensions must be managed:

- **Automation Bias & Syntactic Fluency:** The model achieves a remarkably low validation perplexity of 1.14. This high fluency creates a dangerous cognitive bias where human analysts assume grammatical perfection correlates with factual truth. As shown in our audits, despite this 1.14 PPL, the model has an 80% failure rate (4/5 categories failed) in copying numerical parameters correctly under greedy search. It is ethically critical to treat all generative outputs as unverified drafts.
- **Deskilling and Over-reliance:** Excessive reliance on automated reports can lead to analyst deskilling, reducing the capability of junior analysts to interpret raw logs manually. Furthermore, a false "Normal/Benign" classification of a novel exploit vector can lead to delayed triage. System boundaries must mandate a Human-in-the-Loop (HITL) sign-off workflow.
- **Data Confidentiality & Leakage:** Causal models trained on proprietary enterprise logs are vulnerable to reconstruction attacks. All training pipelines must apply differential privacy filters, strictly strip internal IP addresses and PII, and run on localized, containerized architectures to prevent leakage to external commercial APIs.

Interpretation for a Non-Technical Audience

In corporate cybersecurity operations, monitoring tools generate millions of complex raw log entries daily. Security analysts face severe alert fatigue when forced to manually read cryptic text codes, leading to operational burnout and delayed threat response. Our fine-tuned generative AI model serves as an automated digital translation assistant:

- **Instant Telemetry Translation:** Converts technical server logs (e.g., Flag: S0, Serror: 0.95) into clear executive threat briefings within 0.2 seconds.
- **80% Reduction in Triage Workload:** Automates initial incident report drafting, allowing tier-1 analysts to focus immediately on active threat containment.
- **Standardized Threat Reporting:** Ensures consistent, high-quality incident documentation across enterprise security teams.

Reproducibility & Environment Setup

Environment dependencies were strictly managed using pip freeze > requirements.txt (torch>=2.0.0, transformers>=4.36.0, datasets, matplotlib, seaborn). Executing generative_model.ipynb top-to-bottom recreates identical fine-tuning runs, evaluation metrics, and figure plots. Fine-tuned model weights and tokenizer configurations were exported to fine_tuned_distilgpt2_security/ (containing model.safetensors, config.json, generation_config.json, tokenizer_config.json, and vocabulary maps) for immediate deployment.

References

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